Reg. No. [ ] [12IS663/13IS663]

# SIXTH SEMESTER B.E/B.TECH EXAMINATION, JUNE – 2016

## DATA BASE INTERNALS

Time: 3 Hours Maximum Marks:70

**Instructions:**

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1. a. Discuss the three main alternatives for what to store as a data entry in an index. **07**

   b. Explain with an appropriate example on how data is organized in a tree based index. **07**

2. a. Discuss on how the disks are called as RAIDS. Explain briefly the different levels of Redundancy. **10**

   b. Differentiate on Fixed length and variable length records. **04**

3. a. Discuss with an appropriate diagram that illustrates the ISAM data structure. **04**

   b. Write an appropriate algorithm to insert an element and also search a key element in a B+ tree of order n. **10**

4. a. Illustrate the use of static hashing over a file consisting of buckets 0 through N-1 with one primary page per bucket initially. **07**

   b. Discuss how linear hashing is used as a dynamic hashing technique with its use of buckets during a rounds of splitting. **07**

5. a. Briefly discuss with an appropriate example for each of the 3 common techniques that are used to develop algorithms for each operator. **06**

   b. Consider the schema with sailors relations: Sailors (sid:integer, Snane: string, rating, integer, age: real) **08**

   Assume that each tuple of sailors is 50 bytes long, that a page can hold 80 sailors tuples, and that we have 500 pages of such tuples. For each of the following selection conditions, estimate the number of pages retrieved, given the catalog information,

   i. Assume that we have a B+trees index T on the search key (sailors, sid, and assume that IHeight(T)=4, INages(T)=50, Low(T)=1 and High(T)=100,000. (a) $\sigma\ \text{sailors}_{e-\text{sid}} = 50,000\,(\text{sailors})$

   ii. Assume that we have a hash index T on the small key (sailors, sid) and assume that IHeight(T)=2, INPages(T)=50, Low(T)=1 and High (T)=1000,000

   (a) $\sigma\ \text{sailors.sid} < 50,000.(\text{sailors})$.

6. a. Mention the two important modifications to the two way merge sort algorithm. Also discuss the clustered B+tree for sorting. 08

b. Write an appropriate algorithm for a hash join. 06

7. a. Discuss the guidelines for index selection. 08

b. Illustrate an suitable query as an example of co-clustering two relations. 06

8. Write short notes on: 14

i. Denormalization

ii. Choices in tuning queries and views